

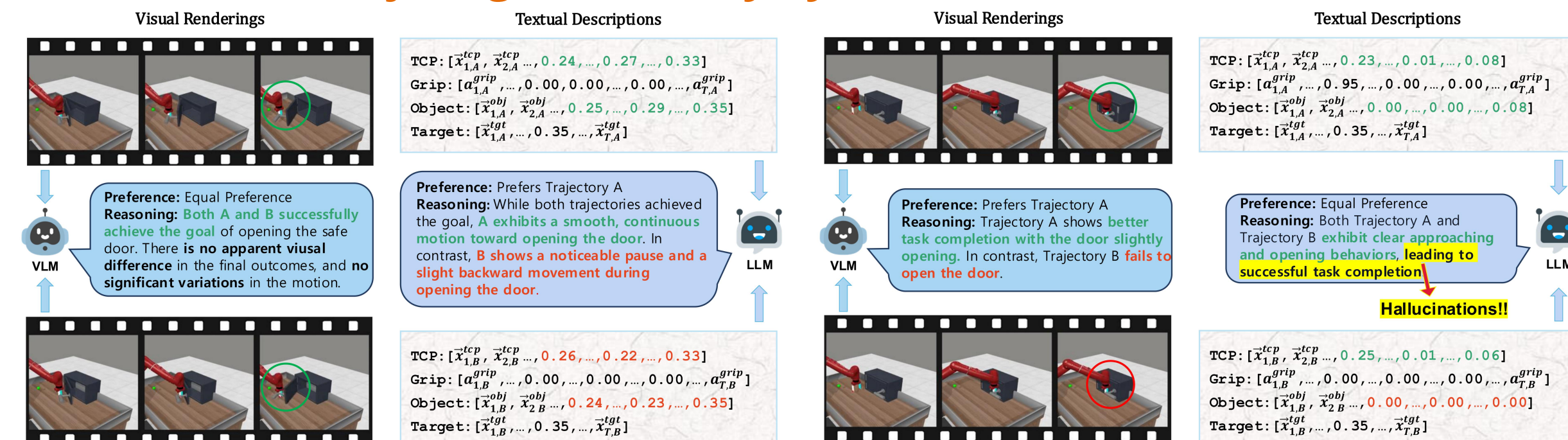
Motivation

Preference-based RL offers an alternative to hand-designed rewards by learning from comparative feedback; yet its **reliance on human labels** prevents scalability.

Our goal: enable scalable, zero-shot PbRL with foundation models (FMs)

Challenges to solve:

1. Limitations of Single-Modality Synthetic Feedback

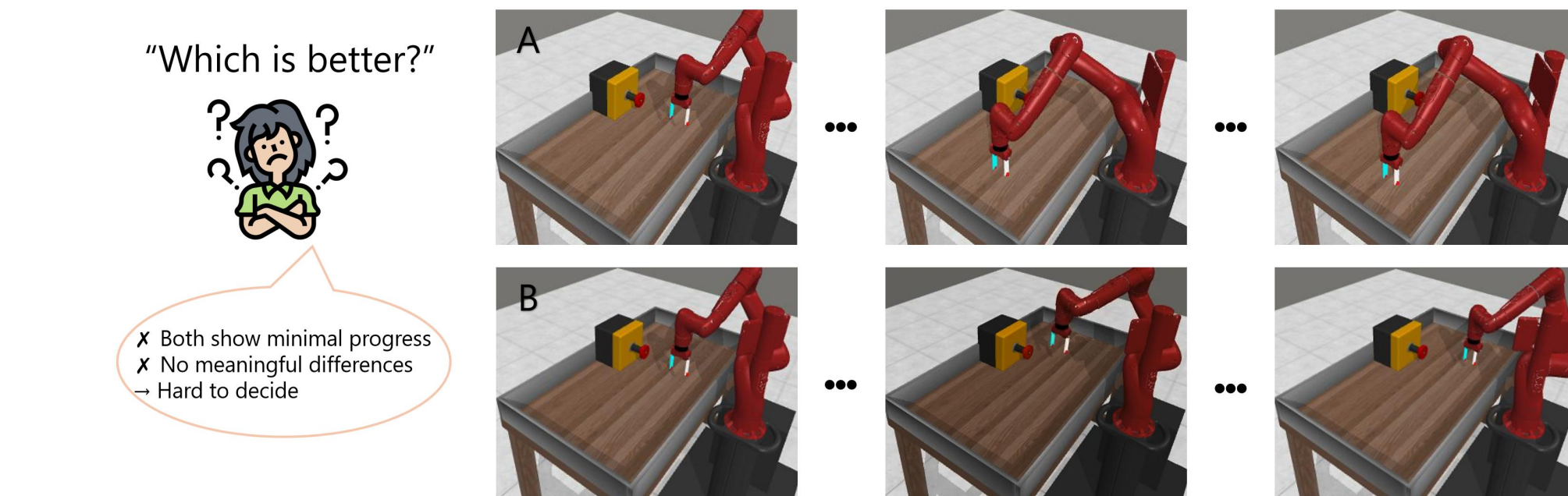


LLMs-only → good temporal and logical reasoning, but often hallucinate or miss fine-grained spatial interactions and key events.

VLMs-only → reliable spatial grounding and goal-state assessment, but limited ability to interpret temporal progression or subtle motion dynamics.

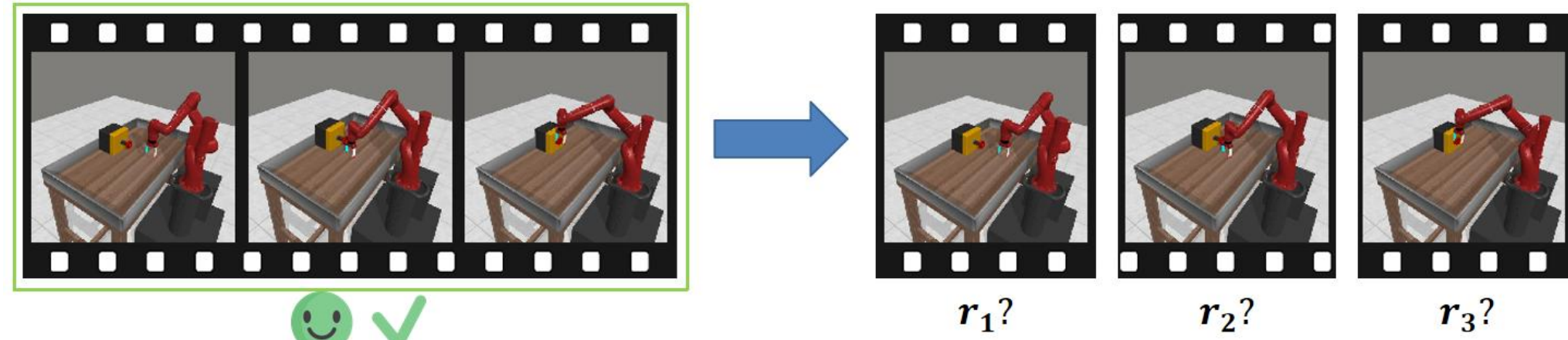
2. Query Ambiguity in Early Training

Early trajectories from random policies are uniformly low quality lacking meaningful task variations → **cannot provide informative comparisons.**

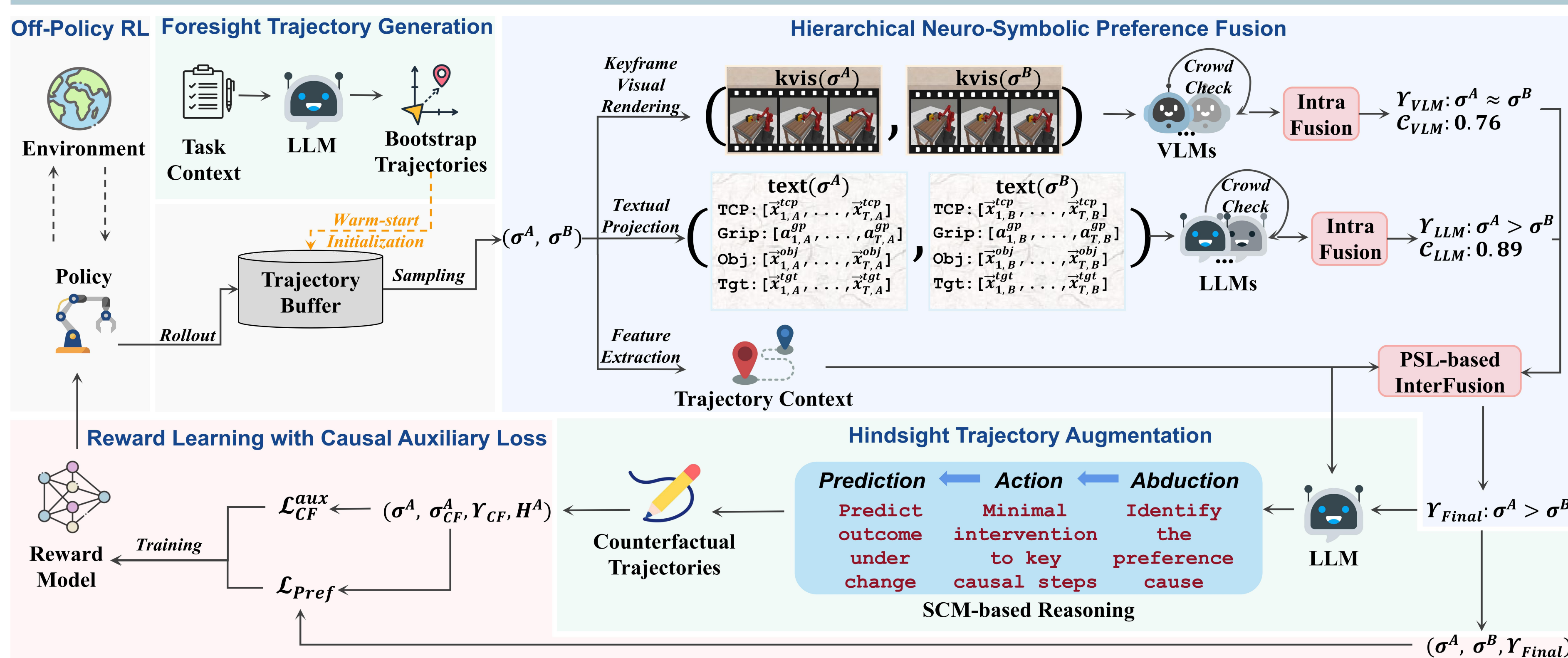


3. Preference Credit Assignment Uncertainty

Preferences are given at trajectory level, but reward models operate at state-action level → **hard to determine which steps caused the preference.**



Method



1. Hierarchical Neuro-Symbolic Multimodal Preference Fusion

Intra-modal Fusion:

- Crowd LLM/VLM evaluations
- Major voting + Confidence estimation

Inter-modal Fusion:

- Fuse VLM and LLM preferences via Probabilistic Soft Logic
- Model confidence, conflicts, agreement, and traj contexts

$$\forall Y, M: \text{IsAgree}(Y) \wedge \text{ConfHigh}(M) \rightarrow \text{FinalLabel}(Y)$$

$$\forall Y: \neg \text{IsAgree}(Y) \wedge \text{VLMLabel}(Y) \wedge \text{ConfHigh}(\text{VLM}) \wedge \text{VDHigh} \rightarrow \text{FinalLabel}(Y)$$

$$\forall Y: \neg \text{IsAgree}(Y) \wedge \text{LLMLabel}(Y) \wedge \text{ConfHigh}(\text{LLM}) \wedge \text{TDHigh} \rightarrow \text{FinalLabel}(Y)$$

$$\neg \text{ConfHigh}(\text{VLM}) \wedge \neg \text{ConfHigh}(\text{LLM}) \rightarrow \text{FinalLabel}(-1)$$

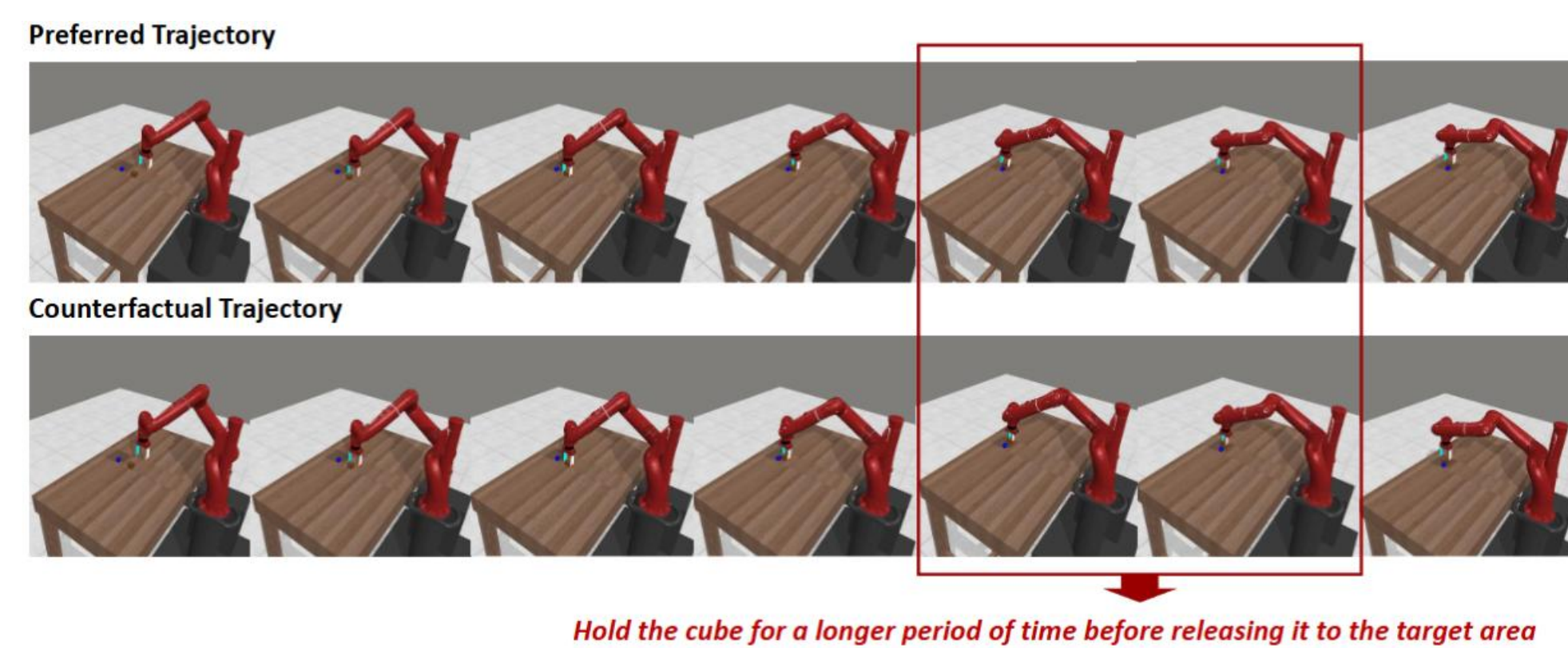
2. Foresight Trajectory Generation

- Generate diverse, task-aligned bootstrapped trajectories
- Code-generation paradigm improves physical coherence
- Use as informative anchors to reduce early query ambiguity

3. Hindsight Counterfactual Augmentation

“What minimal change would make this trajectory less preferred?”

- Minimal step-level edits to reverse preference:



- Train with Auxiliary Causal Loss:

$$\mathcal{L}_{cf}^{aux} = \sum_{t=1}^T H_t \cdot \log \left(1 + \exp \left(r_{\psi}(s_t^{cf}) - r_{\psi}(s_t^*) \right) \right) + \sum_{t=1}^T (1 - H_t) \cdot \left\| r_{\psi}(s_t^*) - r_{\psi}(s_t^{cf}) \right\|_2^2$$

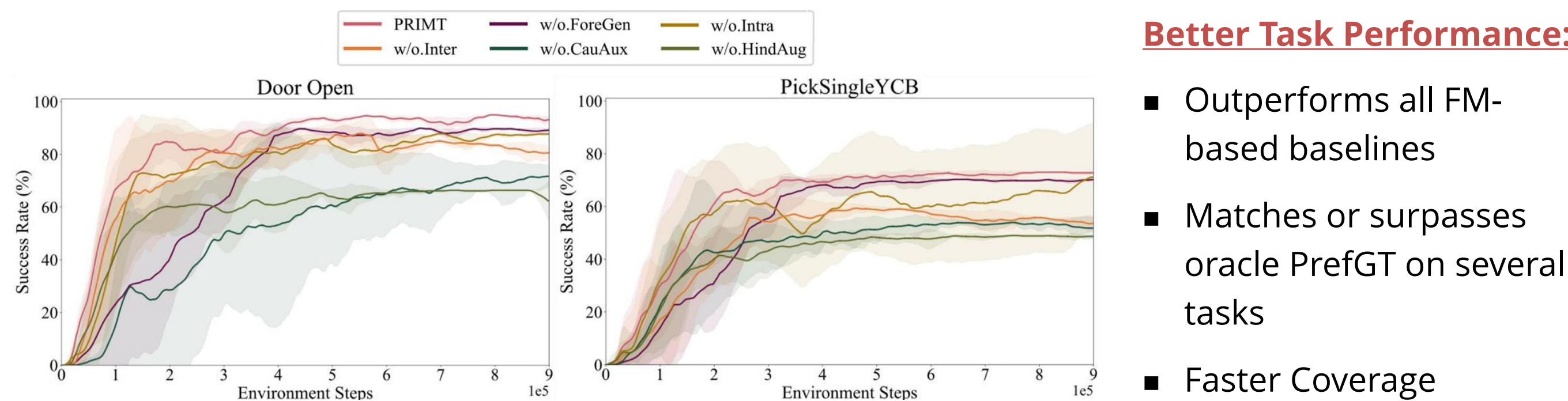
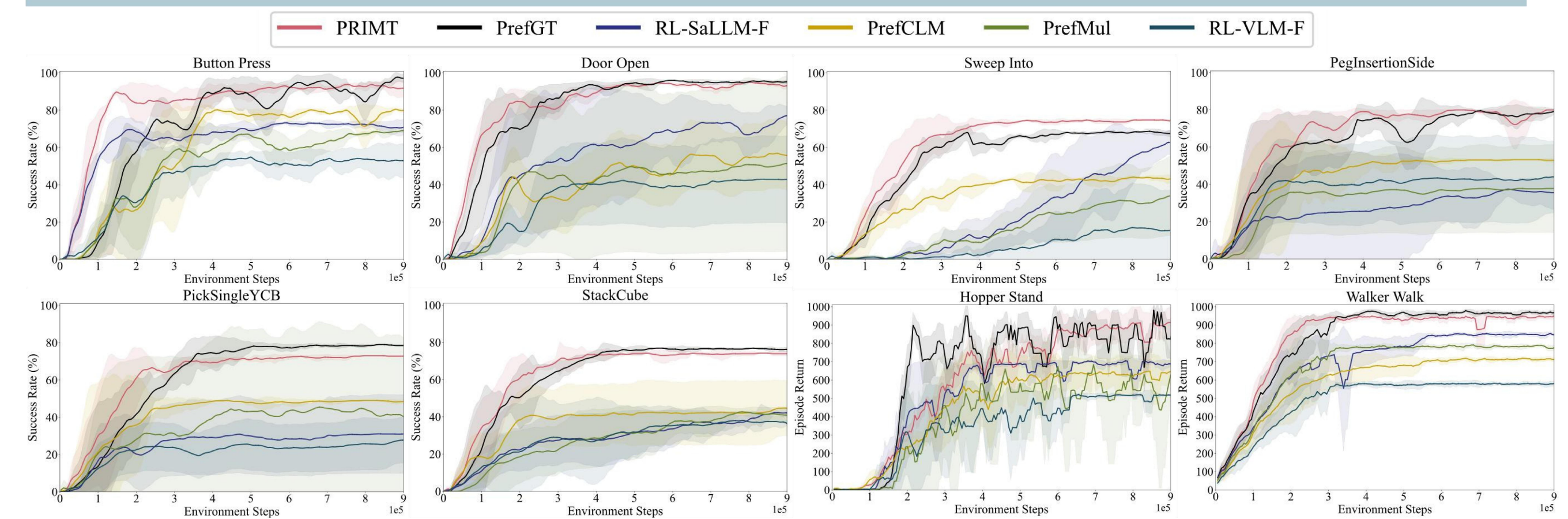
i) causal contrast loss

i) Contrast rewards at edited (causal) steps

ii) reward consistency loss

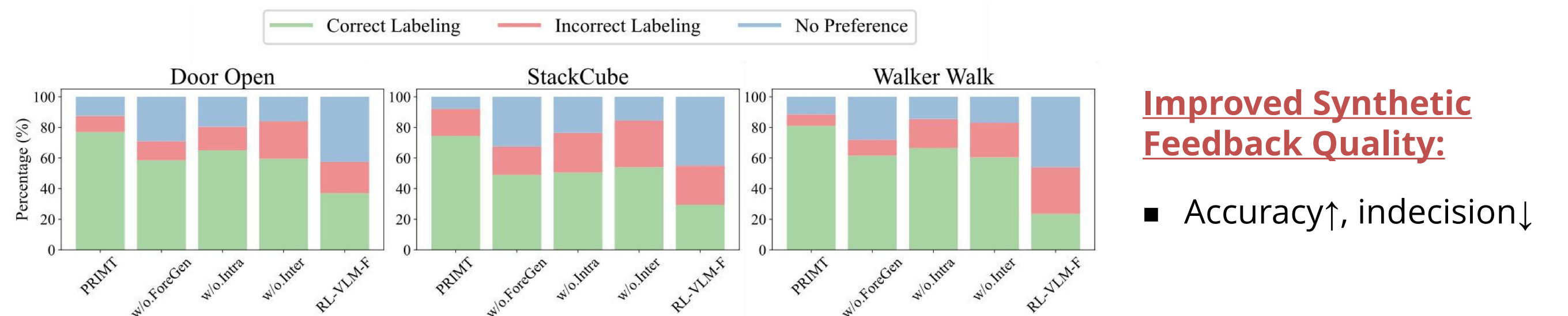
ii) Enforce consistency on unedited regions

Experimental Results



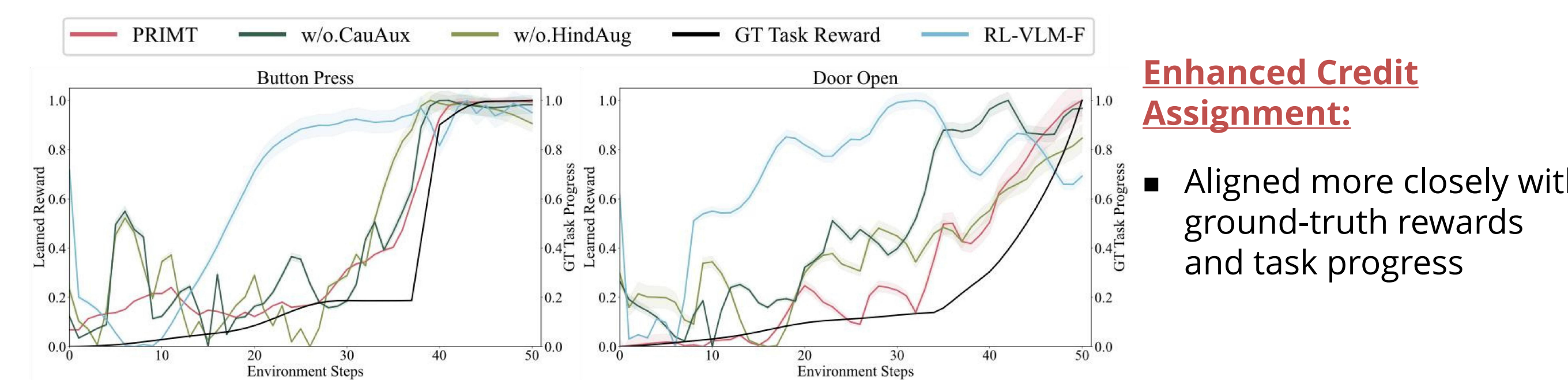
Better Task Performance:

- Outperforms all FM-based baselines
- Matches or surpasses oracle PrefGT on several tasks
- Faster Coverage



Improved Synthetic Feedback Quality:

- Accuracy↑, indecision↓



Enhanced Credit Assignment:

- Aligned more closely with ground-truth rewards and task progress

Baseline	Cost	Time	Performance	Efficiency [†]
		MetaWorld / ManiSkill / DMC		
vs RL-VLM-F	+43%/+39%/+47%	+58%/+43%/+69%	+95%/+117%/+68%	2.0×
vs Sa-LLM-F	+45%/+44%/+38%	+31%/+30%/+46%	+32%/+109%/+19%	1.4×
vs Human (PrefGT)	−92%/−95%/−92%	—/—/—	−1%/−%/−%	47×

- [†] Efficiency = Average performance gain / Average resource increase
- Performance is measured using the final return from the learning curves presented in Figure 2
- Values shown are relative changes across MetaWorld / ManiSkill / DMC environments, respectively

Better Cost-Performance Efficiency